AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 29 · STUDENT LESSON

Uncertainty and failure analysis

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

Last week a model was correct on two constructed validation examples. Why did that not prove it would work reliably on new examples?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Distinguish a predicted class, confidence score and observed correctness. Try the recall before opening notes.

Your learning target

I can distinguish a predicted class, **confidence score** and observed correctness.

I can calculate accuracy and **coverage** for a review policy.

I can investigate failure patterns without claiming an untested cause.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Transfer learning. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Embeddings and neighbourhoods. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from **Training** a small image model. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Distinguish a predicted class, **confidence score** and observed correctness

Calculate accuracy and **coverage** for a review policy

Investigate failure patterns without claiming an untested cause

Retrieve what you already know

> Last week a model was correct on two constructed validation examples. Why did that not prove it would work reliably on new examples?

My starting explanation

This week's end goal

By the end, I can explain uncertainty and failure analysis using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Uncertainty and failure analysis case. Separate observations, given facts and assumptions before choosing an explanation.

A model can be confidently wrong. Use these supplied **output**s for a fictional image classifier: class 1 means a triangle is present; class 0 means it is absent.

The values p1 are probability estimates for class 1, invented for this exercise rather than produced by a trained model. Predict 1 when p1 >= 0.5 and 0 otherwise. The selected-class confidence is max(p1, 1 - p1).

Case and image	p1	Reference label
A clear	0.95	1
B blurred	0.90	0
C clear	0.80	1
D blurred	0.60	0
E clear	0.45	0
F blurred	0.10	1

A predicts 1 with confidence 0.95 and is correct. F predicts 0 with confidence 0.90 and is wrong: p1 = 0.10 is not low confidence in the selected class.

Reference labels are supplied for this exercise; real investigations must check their quality too.

A score alone cannot establish reliability. **Calibration** asks whether, across many comparable cases, **prediction**s assigned a probability such as 0.90 are correct about 90 percent of the time.

One correct or incorrect case cannot answer that question.

List the given facts and one fact that is not supplied.

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Uncertainty and failure analysis. Connect each labelled input, deciding step and result.

The **prediction**s for A through F are 1, 1, 1, 1, 0, 0. A, C and E are correct, so overall accuracy is 3 / 6 = 50 percent.

There are two false positives (B and D), one false negative (F), two true positives and one true negative. Different errors can matter differently; a single accuracy number hides that distinction.

```
rows = [(0.95, 1), (0.90, 0), (0.80, 1),
        (0.60, 0), (0.45, 0), (0.10, 1)]

def report(rows, cutoff):
    if not rows or not 0.5 <= cutoff <= 1:
        raise ValueError("Need rows and cutoff in [0.5, 1]")
    correct = accepted = accepted_correct = 0
    for p1, label in rows:
        if not 0 <= p1 <= 1 or label not in (0, 1):
            raise ValueError("Invalid score or label")
        pred = int(p1 >= 0.5)
        ok = int(pred == label)
        correct += ok
        if max(p1, 1 - p1) >= cutoff:
            accepted += 1
            accepted_correct += ok
    return (correct / len(rows), accepted / len(rows),
            accepted_correct / accepted if accepted else None)
```

```
print(report(rows, 0.8))  # (0.5, 4/6, 0.5) approximately
```

The function returns overall accuracy, **coverage** and accuracy among accepted cases. It assumes numeric scores and binary labels. None means no cases were accepted, so accepted-case accuracy is undefined, not zero or perfect.

Keep counts with proportions and declare the comparison at the boundary.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Uncertainty and failure analysis. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

For all six cases, list false positives and false negatives, count true positives and true negatives, and calculate overall accuracy. Why might a single accuracy hide useful information?

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Raise the acceptance cutoff to 0.85.

Predicted result

Prediction: Only A is accepted: accuracy 0/1, **coverage** 1/3.

Why the result follows

Reason: The most confident supplied case is wrong; a stricter cutoff is not guaranteed to improve this set's accuracy.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Uncertainty and failure analysis to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Confidence score	The model's score for its selected class, which may not match observed reliability
Calibration	Agreement between predicted probabilities and observed frequencies across many comparable cases
Abstention	Withholding an automatic prediction and following a declared alternative process

Word	Meaning
Coverage	The fraction of all cases on which a system makes an automatic prediction

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

Follow the worked trace for Uncertainty and failure analysis, predict a step, then compare with its explanation.

Confidence does not remove every error

CASES	AT CUTOFF 0.80	OBSERVED RESULT
A, B, C, F	Accept automatically	2 correct, 2 wrong
D, E	Defer for review	1 correct, 1 wrong

The policy accepts confidence at least 0.80. Coverage is 4 / 6 and accepted-case accuracy is 2 / 4. The confident errors B and F remain.

Worked explanation · Uncertainty and failure analysis

Deferred cases are not automatically repaired. A review workflow needs an assigned reviewer, suitable **evidence**, a decision record and an outcome check.

Report review workload and the final system's results separately; we have no reviewer outcomes here. Do not erase deferred cases from the report or count them as correct.

Raise the cutoff to 0.95 for the same six cases. Which cases are accepted, what are coverage and accepted-case accuracy, and why does that result not guarantee perfect future predictions?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Uncertainty and failure analysis

Keep positive-class probability separate from confidence in the chosen label.

The facts we will use

Supplied facts

Supplied p(class 1): A: 0.9 reference 0; B: 0.6 reference 1; C: 0.2 reference 0. Predict class 1 at p>=0.5; confidence=max(p, 1-p).

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

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Positive probability is not always chosen-label confidence

	p(class 1)	prediction	confidence	reference
A	0.9	1	0.9	0
B	0.6	1	0.6	1
C	0.2	0	0.8	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/29

Uncertainty and failure analysis

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Predictions

A predicts 1 wrong; B predicts 1 correct; C predicts 0 correct.



2. Selected-label confidence

Confidences are 0.9, 0.6, 0.8. C's confidence is 0.8 despite p(class 1)=0.2.



3. Selective review

Accept confidence>=0.8: A and C; accuracy 1/2 and coverage 2/3.

CHANGE ONE THING

Raise the acceptance cutoff to 0.85.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Predictions	A predicts 1 wrong; B predicts 1 correct; C predicts 0 correct.
2. Selected-label confidence	Confidences are 0.9, 0.6, 0.8. C's confidence is 0.8 despite p(class 1)=0.2.
3. Selective review	Accept confidence>=0.8: A and C; accuracy 1/2 and coverage 2/3.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Positive-class probability and confidence in the chosen label differ when the predicted class is zero. For C, a class-one probability of 0.2 gives class-zero confidence 0.8 under this toy rule.

Selective accuracy counts only accepted cases; **coverage** counts how many were accepted. A stricter cutoff can retain a confidently wrong case and lower accuracy on this supplied set.

What this example does not establish

Keep the stated limit

Supplied scores are not proven calibrated. Review policies need representative held-out

evidence and clear handling of rejected cases.

Keep the stated limit

Read the labelled picture: What is C's positive-class probability and chosen-label confidence? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Raise the acceptance cutoff to 0.85. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Uncertainty and failure analysis claim and explain a repair using the supplied evidence.

Check the claim against the facts

All three clear images are correct; all three blurred images are wrong in this constructed set.

Blur is a candidate explanation, not an established cause: other differences, reference-label errors or pre**process**ing bugs could be involved.

Record case ID, **input** conditions, model and pre**process**ing version, score, **prediction** and checked reference label.

Check the claim against the facts

First inspect B, D and F and check **input** order, resizing and label quality. Then plan new examples that vary blur while keeping the underlying image and label fixed.

Keep related versions of the same source image in one data split; compare both versions within the reserved group. Do not repair cases and report scores on those same cases as independent **evidence**.

Uncertainty can arise from ambiguous **input**s, limited **training evidence** or unfamiliar conditions. A single **confidence score** does not identify the cause and may stay high on unfamiliar **input**s.

Collect **evidence** before selecting a remedy.

Locate the earliest unsupported step

At cutoff 0.80, list accepted and deferred cases, **coverage** and accepted-case accuracy. Can you calculate final accuracy after human review from the supplied data?

What does the report function return for accepted-case accuracy at cutoff 0.99?

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Uncertainty and failure analysis with support, then explain the deciding step in your own words.

For B, E and F, give the predicted class, selected-class confidence and correctness. Explain why p1 = 0.10 does not mean F has low confidence in its predicted class.

> **Raise the cutoff to 0.95 for the same six cases. Which cases are accepted, what are coverage and accepted-case accuracy, and why does that result not guarantee perfect future predictions?**

> **Explain your prediction to a partner before comparing results.**

Practise with less help: Count both types of error

Use workbook W2 for the connected example “Uncertainty and failure analysis”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Uncertainty and failure analysis. Proposed explanation: “A low p1 always means low confidence”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

What is C's positive-class probability and chosen-label confidence? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Uncertainty and failure analysis. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

> For all six cases, list false positives and false negatives, count true positives and true negatives, and calculate overall accuracy. Why might a single accuracy hide useful information?

```
rows = [(0.95, 1), (0.90, 0), (0.80, 1),
        (0.60, 0), (0.45, 0), (0.10, 1)]

def report(rows, cutoff):
    if not rows or not 0.5 <= cutoff <= 1:
        raise ValueError("Need rows and cutoff in [0.5, 1]")
    correct = accepted = accepted_correct = 0
```

```
    for p1, label in rows:
        if not 0 <= p1 <= 1 or label not in (0, 1):
            raise ValueError("Invalid score or label")
        pred = int(p1 >= 0.5)
        ok = int(pred == label)
        correct += ok
        if max(p1, 1 - p1) >= cutoff:
            accepted += 1
            accepted_correct += ok
    return (correct / len(rows), accepted / len(rows),
            accepted_correct / accepted if accepted else None)

print(report(rows, 0.8))  # (0.5, 4/6, 0.5) approximately
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Uncertainty and failure analysis. Explain why each step is justified.

Explain → apply → challenge

For B, E and F, give the predicted class, selected-class confidence and correctness. Explain why p1 = 0.10 does not mean F has low confidence in its predicted class.

> At cutoff 0.80, list accepted and deferred cases, **coverage** and accepted-case accuracy. Can you calculate final accuracy after human review from the supplied data?
>
> What does the report function return for accepted-case accuracy at cutoff 0.99?

All three blurred cases are wrong. A team claims blur caused the errors and plans to put clear images in **training** and their blurred copies in final evaluation.

Identify the unsupported claim and the split problem; propose a better investigation.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled uncertainty and failure analysis **evidence** sheet to Image-pattern Network Investigation. Show how you can distinguish a predicted class, **confidence score** and observed correctness.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Choose review thresholds and model changes using **training** and validation **evidence**, before final evaluation. Report total cases, overall errors, accepted errors, **coverage**, deferred workload and subgroup results.

For tiny groups, give counts and avoid strong claims about future rates. Higher cutoffs do not guarantee higher observed accuracy or safe decisions.

> **A teammate reports “100 percent accurate” after accepting only A at cutoff 0.95. Rewrite the claim honestly and name the evidence still needed before trusting this policy.**

Keep a baseline and record unsuccessful changes too. Next week examines dataset shift: performance can change when the **input**s or relationships encountered in use differ from those seen during development.

All three blurred cases are wrong. A team claims blur caused the errors and plans to put clear images in **training** and their blurred copies in final evaluation.

Identify the unsupported claim and the split problem; propose a better investigation.

> **My limited claim and a useful next check**

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Uncertainty and failure analysis case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Confidence does not remove every error

CASES	AT CUTOFF 0.80	OBSERVED RESULT
A, B, C, F	Accept automatically	2 correct, 2 wrong
D, E	Defer for review	1 correct, 1 wrong

The policy accepts confidence at least 0.80. Coverage is 4 / 6 and accepted-case accuracy is 2 / 4. The confident errors B and F remain.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Analyse new outputs independently

Four new cases have (p1, reference label): K (0.85, 0), L (0.70, 1), M (0.30, 0), N (0.05, 0). Give **prediction**s and confidences, overall accuracy, and accepted cases at cutoff 0.80.

Calculate **coverage** and accepted-case accuracy. Does this policy improve accuracy on the accepted subset here?

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Four new cases have (p1, reference label): K (0.85, 0), L (0.70, 1), M (0.30, 0), N (0.05, 0). Give **prediction**s and confidences, overall accuracy, and accepted cases at cutoff 0.80.

Calculate **coverage** and accepted-case accuracy. Does this policy improve accuracy on the accepted subset here?

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Uncertainty and failure analysis, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Keep positive-class probability separate from confidence in the chosen label.

By the end, I can	How I will show it
Distinguish a predicted class, confidence score and observed correctness	Explain the deciding step, show a trace or test, and state one limit.
Calculate accuracy and coverage for a review policy	Explain the deciding step, show a trace or test, and state one limit.
Investigate failure patterns without claiming an untested cause	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Uncertainty and failure analysis without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Distinguish a predicted class, **confidence score** and observed correctness

> **My deciding step, evidence and one remaining question**
>
> _____
>
> _____
>
> _____
>
> _____

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep positive-class probability separate from confidence in the chosen label.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Accept confidence>=0.8: A and C; accuracy 1/2 and **coverage** 2/3.

1. Selected-label confidence

2. Selective review

3. **Prediction**s

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Selective review Accept confidence>=0.8: A and C; accuracy 1/2 and **coverage** 2/3.

Positive-class probability and confidence in the chosen label differ when the predicted class is zero. For C, a class-one probability of 0.2 gives class-zero confidence 0.8 under this toy rule.

Selective accuracy counts only accepted cases; **coverage** counts how many were accepted. A stricter cutoff can retain a confidently wrong case and lower accuracy on this supplied set.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Raise the acceptance cutoff to 0.85. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Only A is accepted: accuracy 0/1, **coverage** 1/3.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Only A is accepted: accuracy 0/1, **coverage** 1/3. The most confident supplied case is wrong; a stricter cutoff is not guaranteed to improve this set's accuracy.

Repair a missing building block

Review Transfer learning (ai-5/28). Goal: Trace a reused feature extractor and a newly fitted **output** head. Short practice: Which part was trained for the new task?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Embeddings and neighbourhoods (ai-5/26). Goal: Look up a vector using a token ID and preserve sequence shape. Short practice: What would justify claiming nearby vectors mean related topics?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review **Training** a small image model (ai-5/23). Goal: Trace a complete image model with fixed features and one trained weight. Short practice: Does lower **training** loss alone establish that four updates generalise better?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

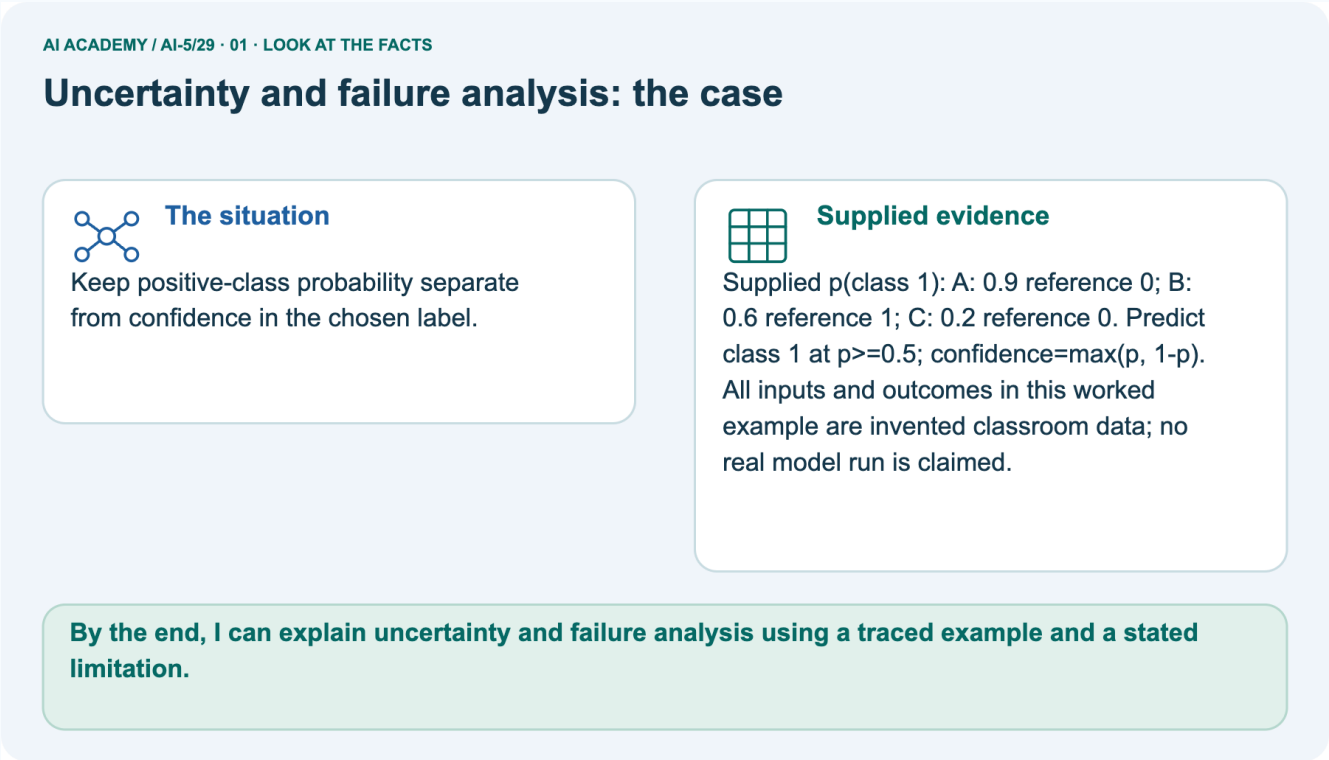
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Uncertainty and failure analysis graphic using its labels, deciding evidence and relevant limitation.

01. Uncertainty and failure analysis: the case



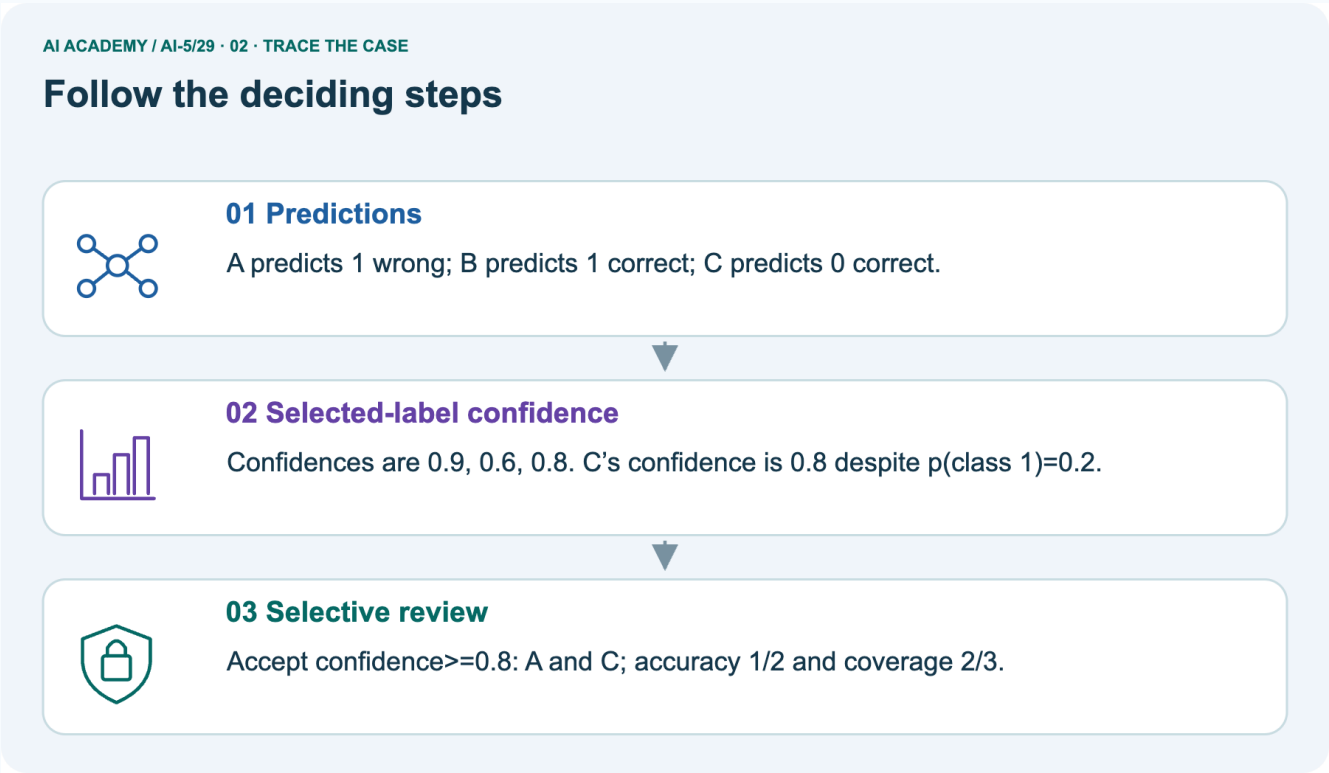
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep positive-class probability separate from confidence in the chosen label. Supplied facts: Supplied p(class 1): A: 0.9 reference 0; B: 0.6 reference 1; C: 0.2 reference 0. Predict class 1 at p>=0.5; confidence=max(p, 1-p).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



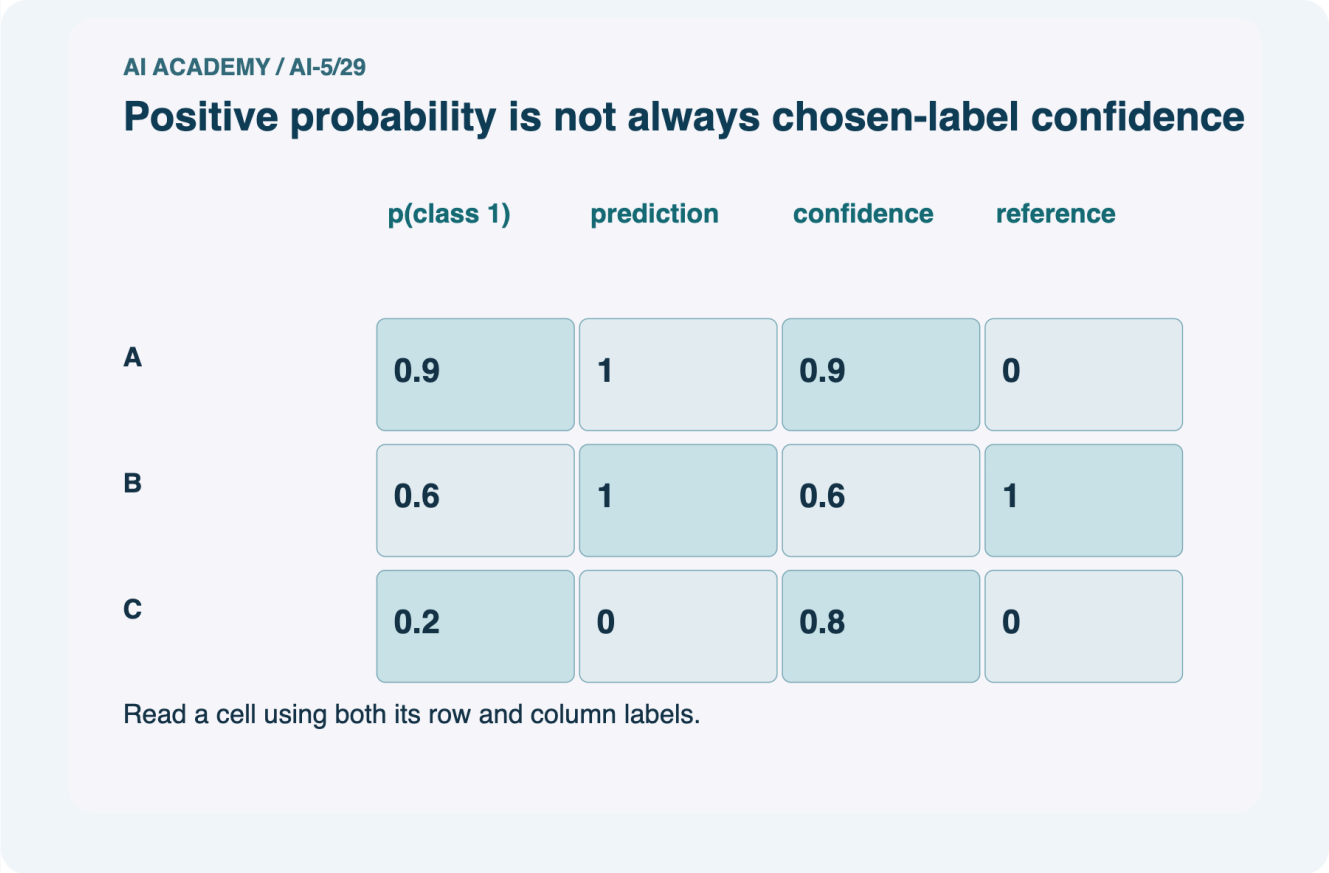
Follow the labelled picture with the matching explanation. Open the vector diagram

Predictions: A predicts 1 wrong; B predicts 1 correct; C predicts 0 correct. → Selected-label confidence: Confidences are 0.9, 0.6, 0.8.

C's confidence is 0.8 despite p(class 1)=0.2. → Selective review: Accept confidence>=0.8: A and C; accuracy 1/2 and **coverage** 2/3.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Positive probability is not always chosen-label confidence



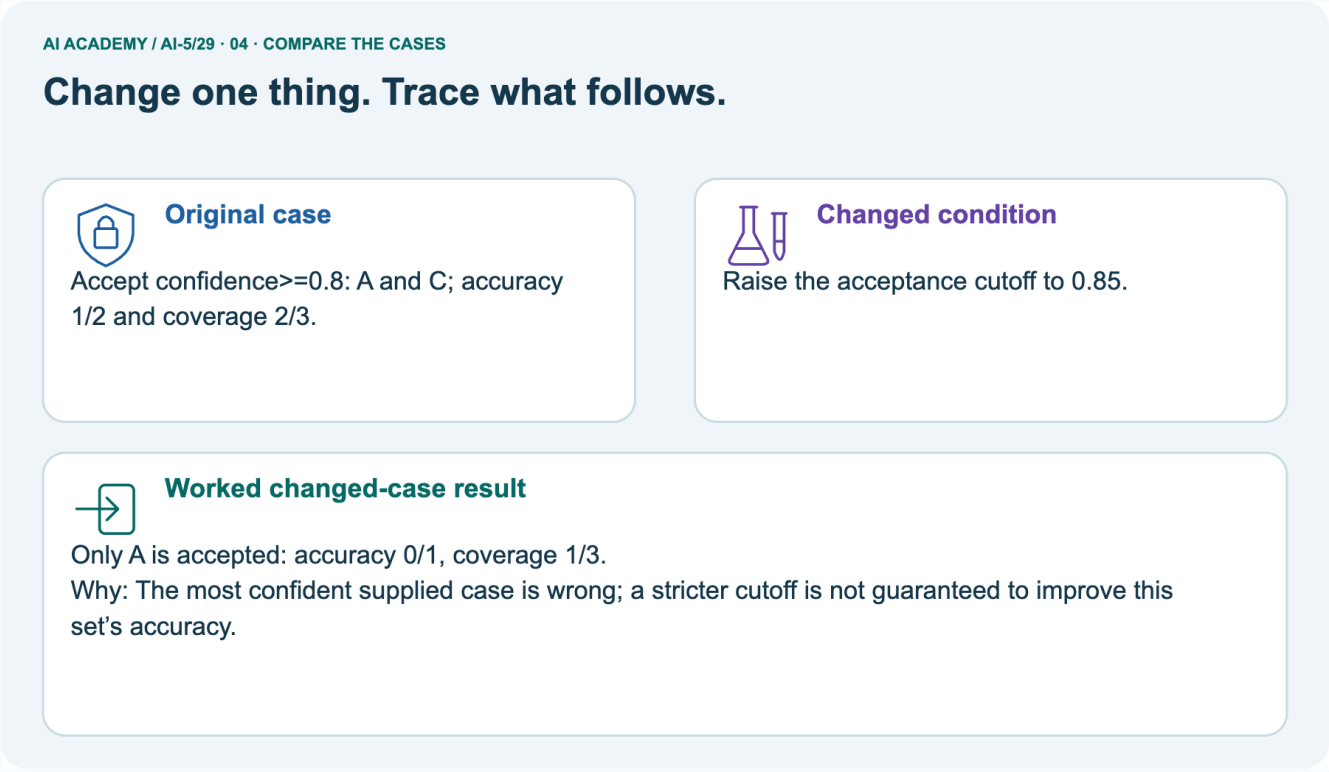
Follow the labelled picture with the matching explanation. Open the vector diagram

Positive probability is not always chosen-label confidence. A: p(class 1) = 0.9, **prediction** = 1, confidence = 0.9, reference = 0. B: p(class 1) = 0.6, **prediction** = 1, confidence = 0.6, reference = 1.

C: p(class 1) = 0.2, **prediction** = 0, confidence = 0.8, reference = 0.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



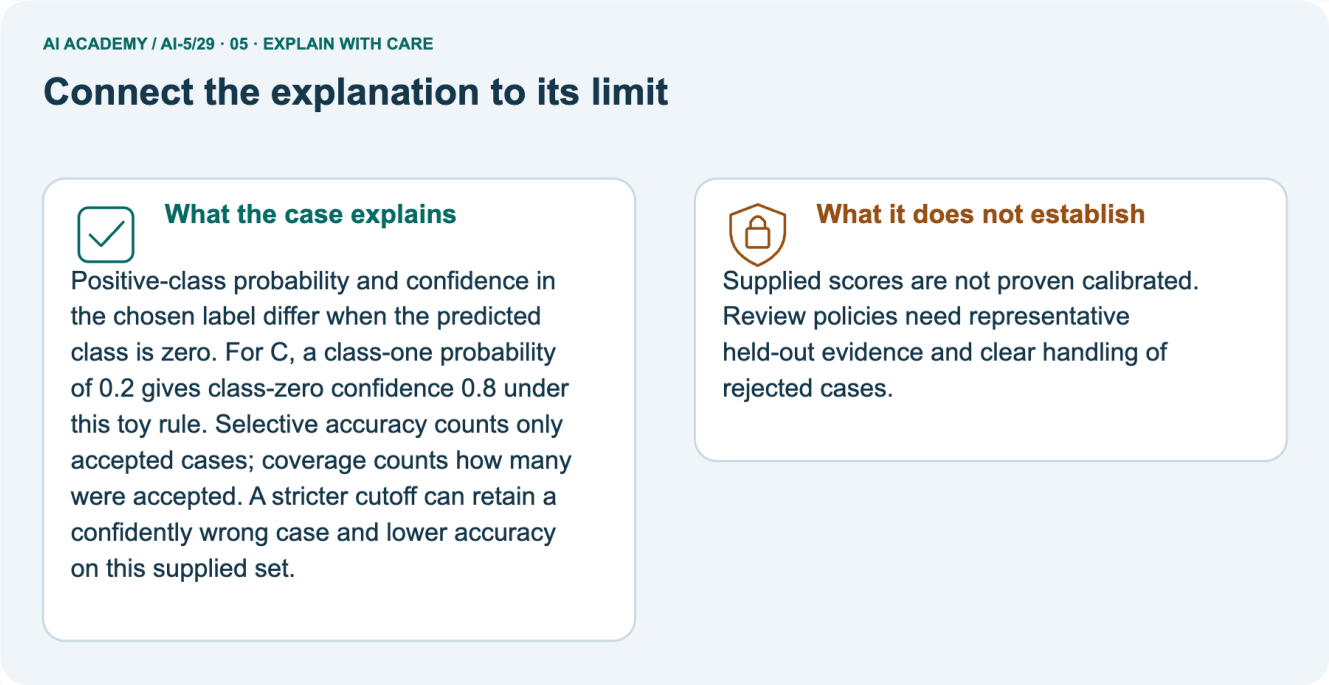
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Accept confidence>=0.8: A and C; accuracy 1/2 and **coverage** 2/3. Changed condition: Raise the acceptance cutoff to 0.85. Expected result: Only A is accepted: accuracy 0/1, **coverage** 1/3.

Reason: The most confident supplied case is wrong; a stricter cutoff is not guaranteed to improve this set's accuracy.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

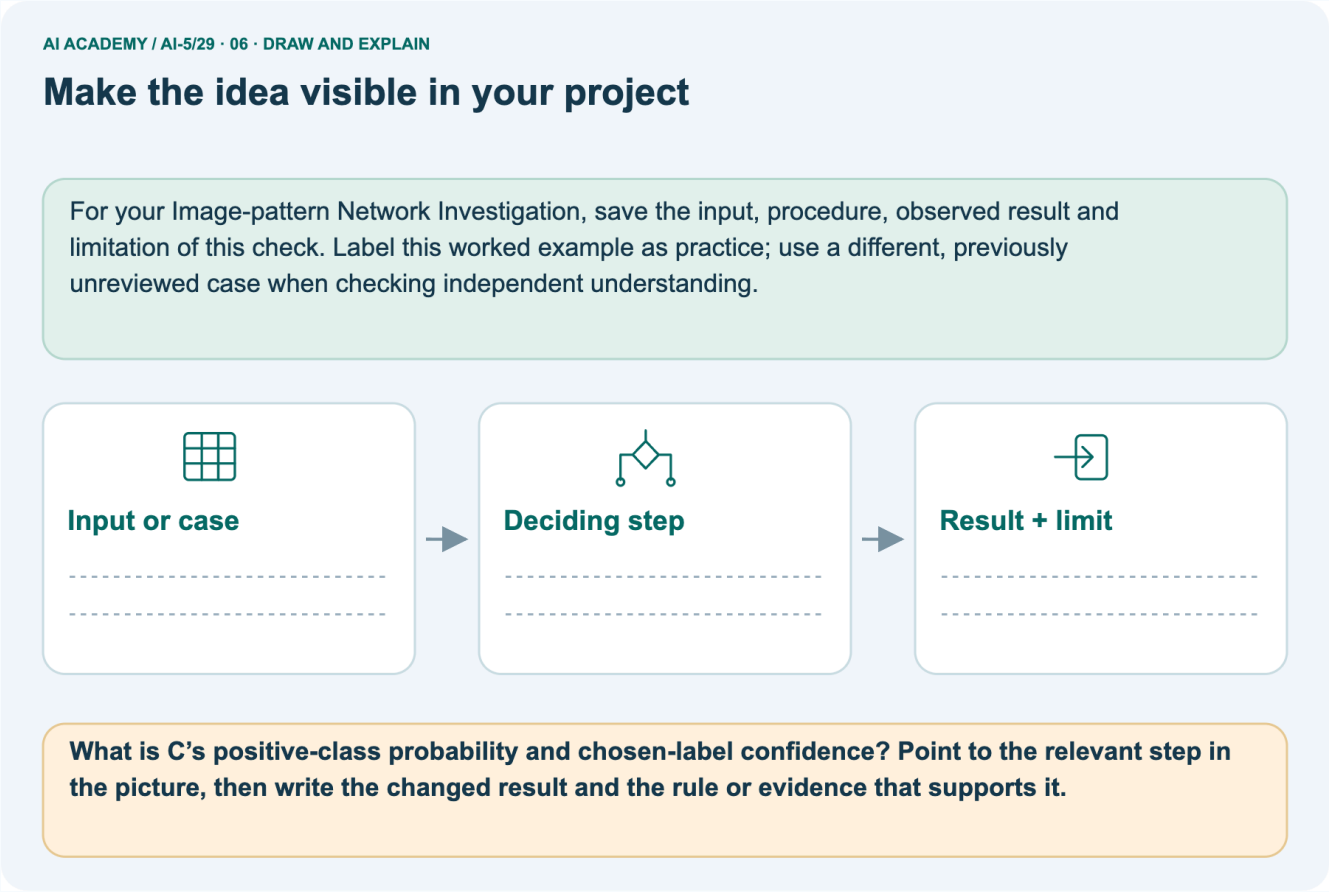
Positive-class probability and confidence in the chosen label differ when the predicted class is zero. For C, a class-one probability of 0.2 gives class-zero confidence 0.8 under this toy rule.

Selective accuracy counts only accepted cases; **coverage** counts how many were accepted. A stricter cutoff can retain a confidently wrong case and lower accuracy on this supplied set.

Limit: Supplied scores are not proven calibrated. Review policies need representative held-out **evidence** and clear handling of rejected cases.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What is C's positive-class probability and chosen-label confidence? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.